

Regional model validation record

Field	Entry
Region	Kiribati (Gilbert Islands), EEZ_941. Regional workbook
Catch source	Sea Around Us regional catch, 1950–2019; landings reference year 2019. Exact Gilbert Islands reporting unit, including unidentified categories; historical download retained. Catch inputs Workbook catch
Selected article	Allain, V.; Nicol, S.; Essington, T.; Okey, T.; Olson, B.; Kirby, D. (2007). An Ecopath with Ecosim model of the Western and Central Pacific Ocean warm pool pelagic ecosystem. WCPFC-SC3-EB SWG/IP-8. DOI: not recorded. Source PDF WCPFC article Paper folder
Other known articles	Griffiths, S.P.; Allain, V.; Hoyle, S.D.; Lawson, T.A.; Nicol, S.J. (2019). Just a FAD? Ecosystem impacts of tuna purse-seine fishing associated with fish aggregating devices in the western Pacific Warm Pool Province. 2005 source and pooled experiment retained for investigation; not selected. DOI 10.1111/fog.12389 Retained article
Selected model	941_20070101_WCPO_Warm_Pool_Option1_(mixed_periods); mixed source periods. Final 31 ecological groups plus one computational import. Pelagic study domain 110–180°E, 15°S–15°N. Selected JSON Model folder Loaded state
Other models from the same article	Initial 24-group WCPO warm-pool model and final 31-group source model, both mixed periods and the same broad domain; final adds stages and forage detail. The adopted D_fixed_M0 option1 is an extraction/parameter variant of the final model. Three Ecosim perturbations are scenarios, not distinct Ecopath inputs. Initial model Final source
Selection rationale	Article: ? — comparative article-selection reason is unrecorded. Model: the recorded selection adopts D_fixed_M0 to preserve juvenile other-mortality flows and improve juvenile production consistency. It remains a surrogate; Griffiths 2019 is deferred for investigation. Recorded rationale
Model extraction	Initial/final table scope is distinguished. Accepted Small BET/YFT P/B and EE differ from the paper. Source YFT/piscivorous diet sums 0.998/1.002 are normalized by the retained loader. GS exceptions are retained; missing BA is derived for 24 groups. Unknown baby/forage catch becomes runtime zero; detritus EE is forced to 1 and synthetic import is added. Native stanza dynamics are not reproduced. Source and runtime review Transformation ledger
SPPR calculation	[MANUAL — Researcher: calculation choices and reasons] [Excluded groups, scope and relevant values; other adjustments]

Field	Entry
GE diagnostics	No negative SPPR entries across the basal-source columns. rho_living = 0.54676655; b = 0.63691435; Detritus (31) SPPR = 2.58026100.
TE diagnostics	No negative SPPR entries across the basal-source columns. rho_living = 0.82430628; Detritus (31) SPPR = 0.25250886.
Geographic fit	Approximate A: 99.7% of the Gilbert Islands EEZ lies in the reported study rectangle. B: about 3.9–4.6% of the study area lies in this EEZ; the range reflects the published-area versus land-excluded coordinate-domain denominator. Broad pelagic overlap does not resolve coastal/reef-fish representativeness. Boundary estimate
Temporal fit	Mixed source inputs: skipjack/forage 1993–2002, yellowfin/bigeye/bycatch 1995–2004 and diet 2001–2007; 2005–2006 assessment editions are not model years. Fixed ecosystem coefficients and composition proxies are transferred across 1950–2019; the 2019 estimate is not an annual ecosystem reconstruction.
Other	The accepted variant retains overall WARN and strict mass balance false (adult BET production residual $\approx 0.728\%$); PP-budget status is OK. Original source/native-model validity and author confirmation remain unestablished. Geographic coverage and nonnegative coefficients do not resolve these limitations. Exact diagnostic evidence
Open issues and next action	[Researcher: open issues and next actions]
Review and reproducibility	Researcher name: [name] review date: [date]

Taxon mapping and coverage

Reference: 2019 landings; all catch labels and source-group scope, including zero catch and unidentified categories.

31 source ecological groups (+1 computational import); 63 catch labels.

Simple-chain PPR totals 167,307,899.00 t C in 2019.

25 labels lack a saved TL/coefficient. All have zero 2019 landings and therefore zero annual contribution; their missing inputs remain unknown.

Method: independent simple trophic chain using saved classic taxon coefficients.

[Taxon mapping appendix](#) and [descriptive Sources sheet](#)

Overall confidence	Taxa	Catch percentage	PPR percentage
High	2	0.0000%	0.0000%
Medium	26	98.5322%	99.5490%
Low	6	0.1778%	0.0341%
Very low	27	1.2901%	0.4170%
Unresolved	2	0.0000%	0.0000%

The summary uses the weaker required membership or allocation confidence. The following tables show each component separately.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Explicit final source membership in the group or stage union.	High	99.3960%
Weak usable taxonomic/functional analogue; habitat/stage mismatch.	Very low	0.3865%
Documented family identity fits the complete source group set.	High	0.0815%
Verified historical/current synonym fits the named source stages.	High	0.0388%
Source lineage partly fits the habitat or stage definition.	Low	0.0341%
Broad catch label uses an assumed represented-pool composition.	Very low	0.0304%
Supported candidate set requires an explicit membership inference.	Medium	0.0293%
Supported ecological extension for an unlisted pelagic piscivore.	Medium	0.0033%
Related small-pelagic taxa extend final listed representatives.	Medium	0.0002%
No meaningful taxonomic/feeding guild or usable analogue.	Unresolved	0.0000%

Membership evidence: [final source definitions and taxonomic/ecological review](#)

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
Historical source landings proportions supply an assumed regional composition; genuine zeros and the accepted infant exception are explicit.	Medium	99.8200%
Incomplete raw source catch rejects the catch proxy; complete biomass proportions assume regional composition/catchability.	Medium	0.1767%
One eligible group receives the entire taxon; this adds no allocation uncertainty.	High	0.0033%
No eligible compartment supplies a numerical allocation.	Unresolved	0.0000%

Each percentage is the simple-chain PPR associated with taxa using that rule. Weights and assumptions:

[candidate/catch/biomass evidence](#)

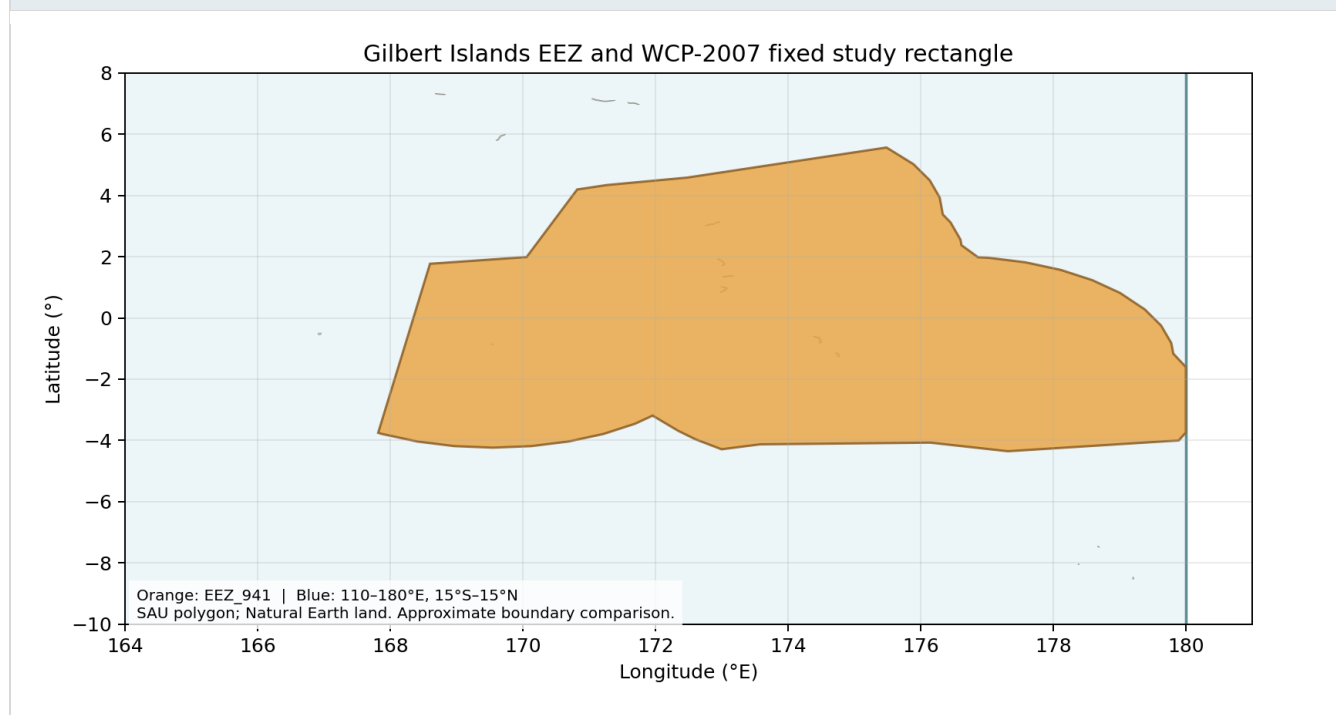
Unresolved: Bivalvia and Echinodermata — no meaningful represented feeding/taxonomic guild; both have zero 2019 catch. Very low: broad marine fish/shark/crustacean/mollusc categories and closest coastal, octopod or unlisted-tuna analogues; all 27 exact labels are flagged in the appendix. They account for 1.2901% of landings and 0.4170% of simple-chain PPR.

Geographic evidence screenshots

Target region R and selected model study area S. Approximate WGS84 boundary comparison with consistent land exclusion; the reported model area gives an alternate B denominator.

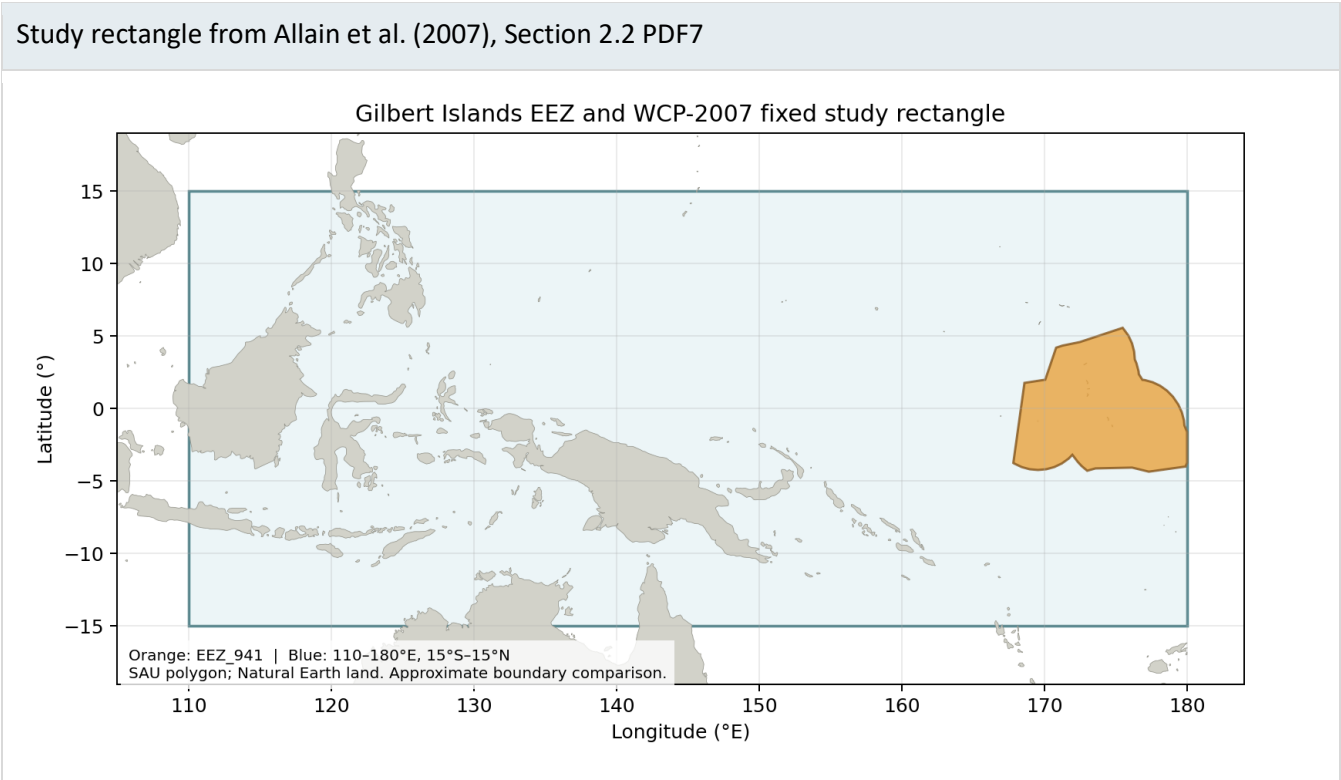
Region area

Kiribati (Gilbert Islands) EEZ_941



Gilbert Islands EEZ boundary: repaired repository Sea Around Us polygon; Natural Earth 1:50m land. [Boundary source](#) | [Image](#)

Study area in the article



Reported bounds: 110–180°E, 15°S–15°N; reconstructed study boundary. Figure 1 PDF3 shows the broader warm-pool/cold-tongue context. [Source PDF7](#) | [Source Figure 1](#) | [Image](#)

